Republic of the Philippines
ENERGY REGULATORY COMMISSION

ENERGY REGULATORY COMMISSION
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RESOLUTION NO. 04, Series of 2021

RULES GOVERNING ACCURACY LIMITS, ACCURACY TESTING AND STANDARDS, AND OTHER REQUIREMENTS FOR THE CERTIFICATION AND TESTING OF METERS AND INSTRUMENT TRANSFORMERS

WHEREAS, pursuant to Section 16(d), (e), (f), and (g) of Commonwealth Act No. 146 (CA 146), the Energy Regulatory Commission (ERC) is mandated to establish the rules and regulations on the standards, classifications, practices, measurements, or service for the measurement of quantity, quality, pressure, initial voltage and other condition pertaining to the accuracy of meters and other auxiliary devices and equipment essential to the metering of electric power and energy;

WHEREAS, Section 43 (t) of Republic Act No. 9136, otherwise known as the *Electric Power Industry Reform Act of 2001 (EPIRA),* empowers the ERC to perform such other regulatory functions as appropriate and necessary to ensure the successful restructuring and modernization of the electric power industry;

WHEREAS, Section 41 of the EPIRA provides that the ERC is primarily responsible in handling consumer complaints and ensuring the adequate promotion of consumer interests;

WHEREAS, on 01 June 2009, the ERC issued Resolution No. 12, Series of 2009, adopting the "Rules and Procedures for the Test and Maintenance of Electric Meters of Distribution Utilities." The said Resolution was published in the "Manila Times" on 26 June 2009 and took effect on 11 July 2009;

WHEREAS, the ERC deems it appropriate to harmonize the accuracy limits, accuracy testing and standards, and other requirements for the certification and testing of meters and instrument transformers across the different pertinent rules and regulations, more specifically, Resolution No. 12, Series of 2009, the Philippine Distribution Code (PDC), and the Philippine Grid Code (PGC);

WHEREAS, the ERC prepared the draft Rules which will govern the accuracy limits, accuracy testing of meters and other requirements of meters and instrument transformer;

WHEREAS, these Rules will guide Distribution Utilities (DUs), Redistributors, Transmission Network Provider (TNP), Generators and all types of customers connected to the distribution system, including the captive customers, contestable customers, and embedded generator-customers toward ensuring the highest level of accuracy in their metering equipment;

WHEREAS, the draft Rules were posted on the ERC website on 20 June 2019 to solicit comments from stakeholders;

WHEREAS, the ERC conducted several public consultations on 04, 05, and 10, July 2019 in Cebu City, Davao City, and Pasig City, respectively; and

WHEREAS, the ERC considered and incorporated the comments of the stakeholders in the proposed Rules.

NOW THEREFORE, **BE IT RESOLVED**, as it is hereby resolved, that the Commission adopts and issues the **RULES GOVERNING ACCURACY LIMITS, ACCURACY TESTING AND STANDARDS, AND OTHER REQUIREMENTS FOR THE CERTIFICATION AND TESTING OF METERS AND INSTRUMENT TRANSFORMERS**, hereto attached as Annex "A" and made integral part of this Resolution.

This Resolution shall take effect fifteen (15) days after its publication in a newspaper of general circulation in the country.

Let copies of this Resolution be furnished the University of the Philippines Law Center – Office of the National Administrative Register (UPLC-ONAR) and all industry stakeholders, as well as published in the ERC website and such other online platform available to the Commission.

Pasig City, 18 February 2021.

ERC
Office of the Chairperson
AVSTD-2021-02-570-0003

AGNES VST DEVANADERA
Chairperson

ON LEAVE
ALEXIS M. LUMBATAN
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RULES GOVERNING ACCURACY LIMITS, ACCURACY TESTING AND STANDARDS, AND OTHER REQUIREMENTS FOR THE CERTIFICATION AND TESTING OF METERS AND INSTRUMENT TRANSFORMERS

ARTICLE I

GENERAL PROVISIONS

Section 1. These Rules shall be known as the ***"Rules Governing Accuracy Limits, Accuracy Testing and Standards, and Other Requirements for the Certification and Testing of Meters and Instrument Transformers."***

Section 2. Objectives - These Rules set the applicable standards and accuracy limits that will be used for all meters and instrument transformers based on the type of customers and their facilities.

Over time, various rules and regulations have been adopted governing the certification and testing of meters and instrument transformers. These Rules aim to provide a reference on the applicable rules and regulations that may apply for the certification and testing of specific meter types and instrument transformers.

These Rules also seek to harmonize the accuracy limits, accuracy testing and standards, and other requirements for the certification and testing of meters and instrument transformers across the different pertinent rules and regulations. More specifically, these are embodied in Resolution No. 12, Series of 2009, or "A Resolution Adopting the Rules and Procedures for the Test and Maintenance of Electric Meters of Distribution Utilities," the Philippine Distribution Code (PDC), and the Philippine Grid Code (PGC).

Section 3. Applicability - These Rules shall apply to the following:

a. All Distribution Utilities (DUs);
b. All types of customers connected to the distribution system, including the: (a) captive customers, (b) contestable customers, and (c) embedded generator-customers;
c. Redistributors;
d. Transmission Network Provider (TNP); and
e. Generators.

Section 4. Definition of Terms

Accuracy Class (Instrument Transformer)	The limits of transformer correction factor in terms of percent error, that have been established to cover

	specific performance ranges for line power factor conditions between 1.0 and 0.6 lag
American National Standards Institute (ANSI)	A private non-profit organization that oversees the development of voluntary consensus standards for products, services, processes, systems, and personnel in the United States
Captive Market	Refers to electricity end-users who do not have the choice of a supplier of electricity, as may be determined by the ERC in accordance with Republic Act No. 9136 or the Electric Power Industry Reform Act of 2001 (EPIRA)
Contestable Market	The electricity end-users who have a choice of a supplier of electricity, as may be determined by the ERC in accordance with the EPIRA
Distribution Utility	Any electric cooperative, private corporation, government-owned utility or existing local government unit which has an exclusive franchise to operate a distribution system in accordance with its franchise and the EPIRA
Embedded Generator	Generating units that are indirectly connected to the grid through the distribution utilities' lines or industrial generating facilities that are synchronized with the grid
EPIRA	Republic Act No. 9136, otherwise known as the Electric Power Industry Reform Act of 2001
Energy Regulatory Commission (ERC)	The independent and quasi-judicial regulatory agency created under Section 38 of the EPIRA
Generation Company	Any person or entity authorized by the ERC to operate facilities used in the generation of electricity

International Electro technical Commission (IEC)	A worldwide organization for Standardization comprising all national electro technical committees
Instrument Transformers	A transformer that reproduces in its secondary circuit, in a definite and known proportion, the voltage or current of its primary circuit with the phase relation substantially preserved
Redistributors	A person or entity, which has an electric service contract with a DU and who re-distributes electricity to other person/s or entity/ies. This may include a building owner, building administrator or homeowners' association charged with the responsibility of redistributing electricity to the individual unit-users of the same building or the owner of the Load-End Power Substation charged with the responsibility of redistributing electricity to the individual building owners within an industrial or commercial complex
Transmission Network Provider (TNP)	The party that is responsible for maintaining adequate Grid capacity in accordance with the provisions of the Philippine Grid Code
Watt-hour meter	A device that measures and records the consumption or production of electricity

ARTICLE II

ACCURACY LIMITS, TESTING STANDARDS, AND OTHER REQUIREMENTS FOR THE CERTIFICATION AND TESTING OF WATT-HOUR METERS

Section 5. General Principles

The accuracy limits, testing standards, and other requirements prescribed under Resolution No. 12, Series of 2009, PDC, and PGC for the certification and testing of watt-hour meters shall be based on the type of metering and its voltage level. Generators, TNP, DUs, and Redistributors shall be guided by the standards and requirements provided under these Rules and other relevant rules and regulations.

Section 6. Pre-Installation Testing/Certification of Watt-hour Meters

All watt-hour meters must be tested, certified, and sealed by the ERC prior to installation on the customers' premises. Their accuracy limits shall be governed by the limits set under Section 7 of these Rules.

Section 7. Watt-hour Meter Accuracy Limits and the Applicable Rules and Regulations

Various rules and regulations have been adopted over time, prescribing watt-hour meter accuracy limits. Table 1 provides for the accuracy limits of watt-hour meters and the applicable rules and regulations governing the subject meter type.

Other provisions of pertinent rules and regulations pertaining to metering point location, measurement, interval data, and other metering equipment, including enclosures, seals, test switch, conduits, wiring provisions, among others, shall continue to be in effect and would need to be complied with by the Generators, TNP, DUs, and Redistributors.

They shall be guided by the following meter accuracy limits for each meter type and the pertinent provisions of existing rules and regulations.

Table 1: Meter Accuracy Limits and Applicable Rules and Regulations

Meter Type	Voltage Level	Accuracy Limits	Applicable Metering Rules and Regulations
Single phase self-contained meters, including 2-element set-up (e.g., Form 1S or Form 2S)	600 Volts and below	The meter shall have an allowable average error within +/- 0.5%	Article V, Resolution No. 12, Series 2009 (excluding Test load points for poly-phase meters)
Three phase self-contained meters and all Instrument Transformer-rated meters	Below 69,000 Volts	The meter shall have an allowable average error within +/- 0.2% for both IEC and ANSI standard meters	a. Chapter 7.2.7 of the 2017 Philippine Distribution Code (PDC); b. IEC 62053.22 2016-11 Edition; c. ANSI C12.20-2015; and d. 2016 Philippine Grid Code (PGC), Chapter 9 GRM 9.2.3.3 Requirements for the Revenue Metering Equipment for WESM-registered Station Service Facilities
Three phase Instrument Transformer-rated meters	69,000 Volts and above	The meter shall have an allowable average error within +/- 0.2% for both IEC and ANSI standard meters	a. Requirements for the Revenue Metering Equipment, Chapter 9 GRM9.2.3.3 of the 2016 Philippine Grid Code (PGC); b. IEC 62053.22 2016-11 Edition; and c. ANSI C12.20-2015

Generation Companies, TNP, and DUs shall be guided accordingly by the requirements and standards provided under these Rules.

Section 8. In-Service Testing Requirements for Watt-hour Meter

All DU meters that are in service should be tested in accordance with the Statistical Sampling Program for In-service Meters, pursuant to Article VI of Resolution No. 12, Series of 2009 otherwise known as the *"Rules and Procedures for the Test and Maintenance of Electric Meters of Distribution Utilities," as*

amended in Resolution No. 21, Series of 2013, and Resolution No 14, Series of 2017.

All meters of Generation Companies and TNP that are in-service should be tested in accordance with the pertinent provisions of the PGC.

ARTICLE III

ACCURACY LIMITS, TESTING, AND OTHER REQUIREMENTS FOR THE CERTIFICATION AND TESTING OF INSTRUMENT TRANSFORMERS

Section 9. General Principles

The accuracy limits, testing, and other requirements for instrument transformers shall be based on its voltage level.

The criteria and other requirements for the certification and testing of instrument transformers shall be guided by the standards and requirements provided under these Rules and other relevant rules and regulations.

Section 10. Testing/Certification of Instrument Transformers

All instrument transformers of DUs must be tested and certified by the ERC prior to installation in the customers' premises.

ERC stickers shall be attached to the instrument transformers by the ERC as a warranty that they are compliant with the standards and requirements under existing rules and regulations, and may, therefore, be used. Their accuracy limits shall be governed by the limits set under Section 11 of these Rules.

All instrument transformers of the Generators and TNP must be tested in accordance with the pertinent provisions of the PGC.

Section 11. Accuracy Limits for Instrument Transformers and Applicable Rules and Regulations

The accuracy class and the corresponding voltage level of an instrument transformer shall be governed by the following rules and regulations, as follows:

Table 2: Instrument Transformer Accuracy Class

Voltage Level	Instrument Transformer Accuracy class	Applicable Rules
Below 69,000 Volts	The instrument transformer shall have an accuracy class of 0.2 or better for Instrument Transformers that conform to IEC standards; or have an accuracy class of 0.3 or better for Instrument Transformers that conform to ANSI standards	a. Chapter 7.2.6 (Requirements for Instrument Transformers) of the Philippine Distribution Code (PDC). b. Chapter 9, GRM9.2.3.1 and GRM9.2.3.2 of the Philippine Grid Code (PGC) Requirements for the Revenue Metering Equipment (for WESM-registered Station Service Facilities)
69,000 Volts and above	The instrument transformer shall have an accuracy class of 0.2 or better for Instrument Transformers that conform to IEC standards; or have an accuracy class of 0.3 or better for Instrument Transformers that conform to ANSI standards	Chapter 9, GRM9.2.3.1 and GRM9.2.3.2 (Requirements for the Revenue Metering Equipment) of the Philippine Grid Code (PGC)

The allowable accuracy limits in conformance with IEC or ANSI standards, for each of the accuracy class of instrument transformers described under Table 2, are as follows:

Table 3a: IEC Standard Accuracy Limit per Instrument Transformer Accuracy Class

Standard	Accuracy Class	Current Transformer			Voltage Transformer
		100% Rated Current	20% Rated Current	5% Rated Current	@ 80% to 120% Rated Voltage
IEC 61869-2 IEC 61869-3	0.1	+/- 0.10%	+/- 0.20%	+/- 0.40%	+/- 0.10%
	0.2	+/- 0.20%	+/- 0.35%	+/- 0.75%	+/- 0.20%

Table 3b: ANSI Standard Accuracy Limits per Instrument Transformer Accuracy Class

Standard	Accuracy Class	Current Transformer			Voltage Transformer
		100% Rated Current	10% Rated Current	5% Rated Current	@ 90% to 110% rated Voltage
ANSI IEEE C57.13.6	0.15	+/- 0.15%	-	+/-0.3%	+/- 0.15%
ANSI IEEE C57.13	0.3	+/- 0.3%	+/-0.6%	-	+/- 0.3%
	0.3S	+/- 0.3%	-	+/- 0.3%	-

Actual modifications must be undertaken by the Generators, TNP, and DUs to comply with the new requirements under these Rules.

Section 12. Secondary Burden

The total burden of each secondary metering circuit, consisting of the burdens coming from all connected devices and secondary cables, shall not exceed the rated burden limit of the instrument transformer.

The instrument transformers shall be connected only to a revenue meter with a burden that does not exceed its rated burden limit in order to ensure the accuracy of the entire metering facility.

Upon installation in service of the instrument transformer, the connection of the leads shall be verified and validated against the polarity markings of the instrument transformer, using an industry-accepted methodology.

Section 13. Commissioning Test Requirements

The Generators, TNP, and DUs shall ensure the proper installation of the instrument transformer in the customers' premises, which should be compliant with the standards and requirements under these Rules and other existing rules and regulations before the same shall be commissioned. They shall conduct a visual inspection to ascertain the proper installation of the instrument transformer.

Such inspection shall include the following: (a) comparing equipment nameplate data with the design specifications; (b) the physical condition of the facilities; (c) equipment cleanliness; (d) grounding; and (e) wiring continuity.

The Commissioning Test to be conducted by the Generators, TNP, and DUs may be witnessed by the customer and/or his/her authorized representatives.

Section 14. In-Service Testing Requirements

All instrument transformers shall remain accurate at all times, while in-service.

All instrument transformers used by Generation Companies and TNP that are in-service should be tested in accordance with Chapter 9 of the PGC.

The Generation Companies, TNP, and DUs shall conduct a visual inspection of the instrument transformers to check for signs of physical damage, discolored terminals that may be due to overloading, change in texture or appearance of the insulation of the bushings, and other critical parts that affect the accuracy of the metering installation. This inspection shall be done based on the maintenance program of Generators, TNP, and DUs.

Instrument transformers with signs of defects or abnormality shall be subjected to accuracy testing by the ERC upon request of either the customer, Generation Company, TNP, or DU. The accuracy test shall be performed in the presence of the customer, Generation Company, TNP, or DU. All in-service instrument transformers found to be defective shall be replaced as soon as practicable and shall be subjected to accuracy tests. While the instrument transformer has not been replaced, the transformer correction factors shall be used for billing purposes subject to adjustment as determined by the ERC based on the result of the accuracy test.

ARTICLE IV

FEES AND CHARGES

Section 15. The existing instrument transformer testing fees approved under Resolution No. 12, Series of 2009 is hereby amended, as follows:

Table 4: Fees and Charges

INSTRUMENT TRANSFORMER TESTING	Fees
ERC Laboratory testing of Instrument Transformer (IT) (Walk-in)	PhP 290.00 per tap
Regular Testing of Instrument Transformer (Pre-Installation)	PhP 290.00 per tap
On-site Spot Testing of Instrument Transformer For Other Entities	Php 1,160.00 per tap

ARTICLE V

FINAL PROVISIONS

Section 16. **Imposition of Fines and Penalties.** – Violation of any of these Rules shall be subject to ERC Rules and Regulations and other existing applicable laws.

Section 17. **Separability Clause.** – If any of the foregoing amendments is declared unconstitutional or invalid, the other provisions which are not affected thereby shall remain in force and effect, unless such declaration would render the whole set of Rules unenforceable or non-implementable.

Section 18. Exception Clause – Where good cause appears, the ERC may allow an exception from any provisions of these Rules if such exception is found to be for the interest of the public and is not contrary to law or any other related rules and regulations.

Section 19. **Repealing Clause.** – All Rules and Guidelines, or portion thereof, issued by the ERC, which are inconsistent with these Rules are hereby repealed or modified accordingly.

Section 20. Effectivity. – These Rules shall take effect fifteen (15) days after its publication in a newspaper of general circulation in the country.

Pasig City, 18 February 2021.

ERC
Office of the Chairperson
AVSTD-2021-02-570-0003

AGNES VST DEVANADERA
Chairperson

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